Deep Learning & Machine Learning in High Throughput Image-Based Yeast Sorting

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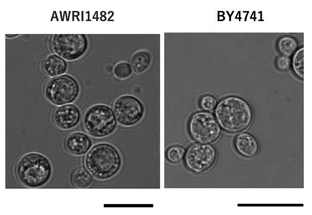
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Abstract

Saccharomyces cerevisiae (Yeast) is arguably one of the best used, most versatile, easily genetically manipulable organism and the very promising candidate for lipid production used in diverse industrial applications. In our study to enhance the efficiency and accuracy of sorting Yeast cells mutants based on cell lipid content and lipid localization, we tested advanced image analysis techniques such as deep learning (DL) and machine learning (ML) which can offer significant advantages over traditional classification methods like feature gating which is limited to classify based on one feature. This study evaluates the effectiveness of DL, ML, and feature gating in optimizing high throughput image-activated sorting processes for mutants classification which can be further used to maximize lipid production by creating better *Saccharomyces cerevisiae* mutant cells. By comparing these AI based methods, we aim to identify the most effective strategy for improving detection accuracy of mutant cells, lipid yield and get the desires quality of *S. cerevisiae,* therefore advancing automated biotechnological applications using Artificial Intelligence.

Confocal Microscopy Images of the two Mutant Yeast strain used, lipids marked with Nilered dye

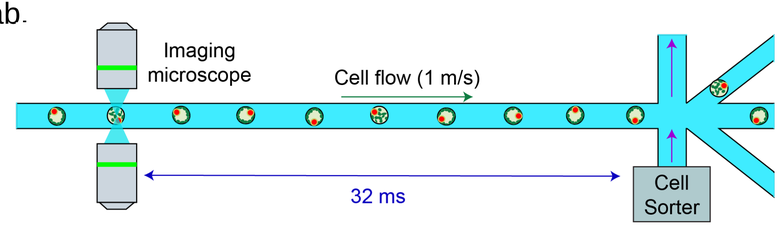


Scale bars: 10µm

AWRI Yeast mutant reported significant higher lipid synthesis. BY strain is the control

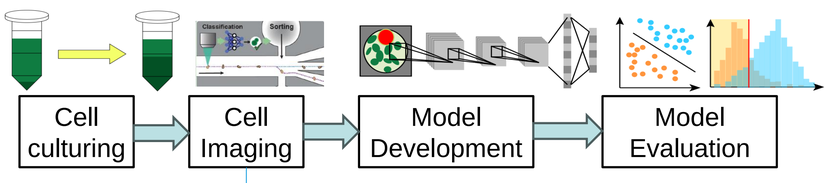
Material

To compare the effectiveness of deep learning (DL), classical machine learning (ML), and feature gating we used AWRI 1482 (Australian Wine Research Institute Strain) and BY 4741(Baker's Yeast Strain) with fat vesicles marked with Nilered dye strains provided by collaboration with Prof. Yoshikazu Ohya's lab. Then single cell images of the cells were taken by the intelligent image activated cell sorting system which is built in house at the Goda Lab.



We wrote code on Python 3.7 by using OpenCV, Scikit-image libraries and VS Code for preprocessing images and extracting features. To test Machine Learning Classifier we used open source library Scikit-Learn. Deep learning models were trained tested and validated using open-source software AIDeveloper.

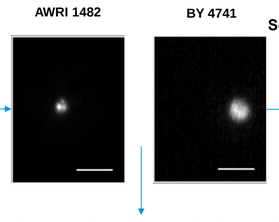
Method



116000+ Raw Images taken from Image flow cytometry system

AWRI 1482 **BY 4741**

Scale bars: 10µm



Different Shape, Size and Intensity based features extracted from each of the cell

Shape Features	Size Features	Intensity Features
Eccentricity	Area	Standard Intensity
Form Factor	Perimeter	Integrated Intensity Edge
Compactness	Bounding Box Area	Maximum Intensity Edge
Extent	Major Axis Length	Minimum Intensity Edge
Solidity	Minor Axis Length	Mass Displacement
	Equivalent Diameter	Lower Quartile Intensity
	Convex Area	Median Intensity
	Maximum Radius	Median Absolute Deviation Intensity
	Median Radius	Upper Quartile Intensity
	Mean Radius	Mean Intensity Edge
	Minimum Feret Diameter	Standard Intensity Edge
	Maximum Feret Diameter	Integrated Intensity

29 Feature Gating, six Machine Learning Classifiers and two Deep Learning Models tested

Feature Gating	Machine Learning Classifers	Convolution Nueral Network(CNN) based Deep Learning Models
Based on all 29 size, shape and Intensity values	K-nearest neighbors (KNN)	DeepCNN-6 Model
	Logistic regression (LR)	DeepCNN-8 Model
	Decision tree (DT)	
	Support vector machine (SVM)	
	Random forest (RF)	
	Histogram-based gradient boosting (HGB)	

Results & Discussions

Calculation of Predictions Accuracy by Confusion Matrix

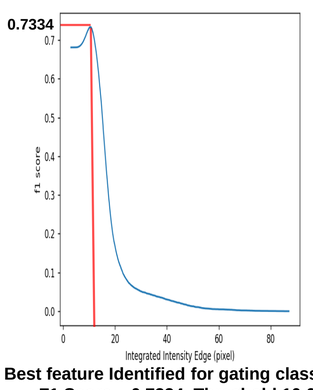
True \ Predicted	*AWRI Mutant*	BY WT
AWRI Mutant	True Positive (TP)	False Negative (FN)
BY WT	False Positive (FP)	True Negative (FN)

Evaluate models with their Calculated F1 scores

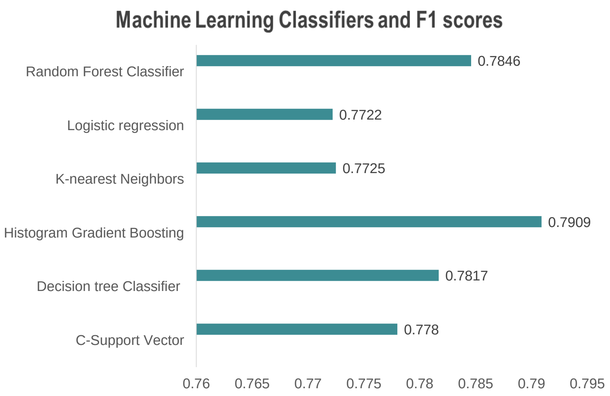
$$Precision = \frac{TP}{TP + FP}$$

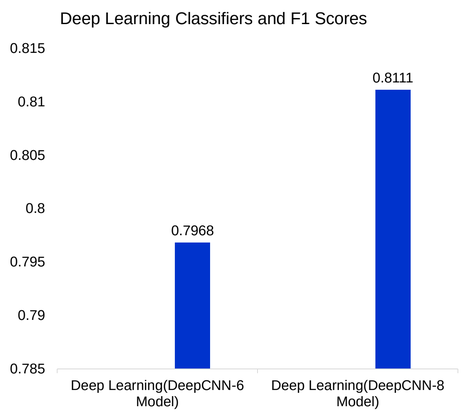
$$Recall = \frac{TP}{TP + FN}$$

$$F1\ score = \frac{2}{Precision^{-1} + Recall^{-1}}$$

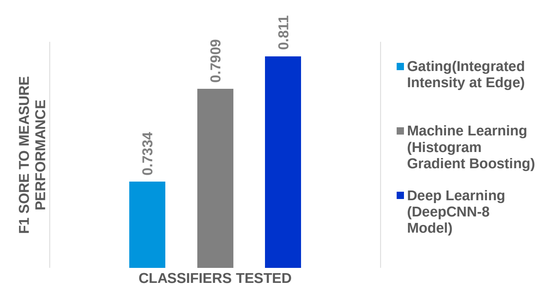


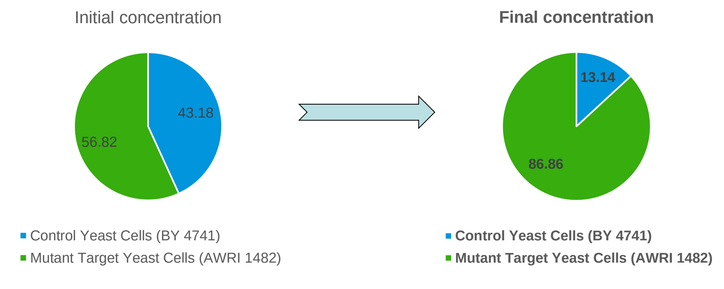
Best feature Identified for gating classifier
F1 Score – 0.7334, Threshold 10.6





Conclusion





Incorporating automated feature selection and analysis technique like machine learning classifiers showed efficient clubbing of these multiple features together and hence presented better F1 scores. However both the advance CNN image analysis models trained in Deep Learning outperformed all six conventional Machine Learning classifiers. This study present a significant improvement in sorting accuracy by using DeepCNN-8 and DeepCNN-6, demonstrating the superiority of AI-driven approach in optimizing lipid production in Saccharomyces cerevisiae. The best classification method to identify mutant in high throughput single cell image based sorting was DeepCNN-8 model with the F1 score of 0.81.

Reference

- Intelligent Image-Activated Cell Sorting. Cell, 175(1), 266-276.e13 doi: 10.1016/j.cell.2018.08.028
- Is AI essential? Examining the need for deep learning in image-activated sorting of Saccharomyces cerevisiae. Lab Chip. 2023 Sep 26;23(19):4232-4244. doi: 10.1039/d3lc00556a.

Future Prospects

I am motivated and inspired by being part of this exemplary research conducted by the fine amalgation of physics, chemistry, electronics, biology and computer science. The future perspectives of this study are wide spread and interesting. In many bioengineering applications obtaining the hard ground truth labels of sufficient quality or quantity to train supervised models is time-consuming, cost-forbidden, or simply unfeasible. I worked and am intrigued by the wonders we can achieve by advancing this serendipity enabling machine by more capable techniques like unsupervised learning and high throughput super resolution imaging such as Structured Illumination Microscopy (SIM), Stimulated Emission Depletion (STED) microscopy. This hold high potential to open doors for much precise classifications, mutant studies, new biomarkers and drug discovery.

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